# FOUNDATION COURSE

START YOUR PREPARATION FROM BASICS

## Prerequisites of Engineering Mathematics

Lecture no. 19

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# Recap *of previous lecture*

## DEFINITE INTEGRATION



# Topics to be covered

"APPLICATION OF LAPLACE TRANSFORM"

(only for EC/EE/IN)



**Defⁿ:** Let $f(t)$ is any funcⁿ the $\int_0^\infty e^{-st} \cdot f(t)\,dt = L.T$ of $f(t)$

$= L\{f(t)\} = F(s)$

i.e. $\boxed{L\{f(t)\} = \int_0^\infty e^{-st} \cdot f(t)\,dt = F(s)}$

$\boxed{\lfloor n+1 = n! \text{ if } n \in I^+}$

$\lfloor 5 = 4!, \lfloor 3 = 2!$

eg $L\{t^2\} = ? = \int_0^\infty e^{-st}(t^2)\,dt = \ldots\ldots = \frac{2}{s^3}$

eg $L\{1\} = ? = \int_0^\infty e^{-st}(1)\,dt = \ldots\ldots = \frac{1}{s}$

g $L\{e^{at}\} = ? = \int_0^\infty e^{-st}(e^{at})\,dt = \int_0^\infty e^{-(s-a)t}\,dt = \left(\frac{e^{-(s-a)}}{-(s-a)}\right)_0^\infty = \frac{1}{s-a}$

5 $L\{\sin t\} = ? = \int_0^\infty e^{-st}(\sin t)\,dt = \ldots\ldots = \frac{1}{s^2+1}$

## 1st Shifting :- if $L\{f(t)\} = F(s)$ then $L\{e^{at} f(t)\} = F(s-a)$.

eg $L\{e^{-2t} \cdot \sin t\} = ?$ $= F(s+2) = \frac{1}{(s+2)^2+1}$

$\because L\{\sin t\} = \frac{1}{s^2+1} = F(s)$

eg $L\{e^{at} \cdot 1\} = ?$ $= F(s-a) = \frac{1}{s-a}$ //

$\because L\{1\} = \frac{1}{s} = F(s)$

(*) $L\{t^n\} = \frac{\Gamma(n+1)}{s^{n+1}}$

$L\{1\} = \frac{\Gamma(0+1)}{s^{0+1}} = \frac{\Gamma 1}{s} = \frac{1}{s}$

$L\{t^1\} = \frac{\Gamma(1+1)}{s^{1+1}} = \frac{\Gamma 2}{s^2} = \frac{1}{s^2}$

$L\{t^2\} = \frac{\Gamma(2+1)}{s^{2+1}} = \frac{\Gamma 3}{s^3} = \frac{2}{s^3}$

eg $L\{t^3\} = \frac{6}{s^4}$ //

| $f(t)$ | $F(s)$ |
|---|---|
| $e^{at}$ | $\frac{1}{s-a}$ |
| $\sin kt$ | $\frac{k}{s^2+k^2}$ |
| $\sinh(kt)$ | $\frac{k}{s^2-k^2}$ |
| $\cos kt$ | $\frac{s}{s^2+k^2}$ |
| $\cosh(kt)$ | $\frac{s}{s^2-k^2}$ |

(*) if $L\{f(t)\} = F(s)$

① $L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$

② $L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s)\, ds.$

③ $L\left\{\int_0^t f(t)\, dt\right\} = \frac{F(s)}{s}$

④ $L\{f'(t)\} = sF(s) - f(0)$

$L\{f''(t)\} = s^2 F(s) - sf(0) - f'(0)$

(*) $L\left\{\frac{\sin at}{t}\right\} = \tan^{-1}\left(\frac{a}{s}\right)$

(*) $\int_0^\infty \left(\frac{\sin at}{t}\right) dt = \frac{\pi}{2}$

# App of L.T in the evaluation of Real Integrals →

**Q:** Evaluate $\int_0^\infty \left(\frac{t^3}{e^{2t}}\right) dt = ? = \int_0^\infty e^{-2t} \cdot t^3 \, dt = ?$

**Sol:**

① W.K.T $L\{t^3\} = \frac{6}{s^4}$ (main step)

② Now using Main Def$^n$ of L.T in LHS only

$$\int_0^\infty e^{-st}(t^3)\, dt = \frac{6}{s^4}$$

③ $s \to 2$

$$\int_0^\infty e^{-2t}(t^3)\, dt = \frac{6}{(2)^4} = \frac{6}{16}$$

**Q:** $\int_0^\infty \left(\frac{\sin t}{t}\right) dt = ?$

**Sol:**

① W.K.T $L\left\{\frac{\sin t}{t}\right\} = \tan^{-1}\left(\frac{1}{s}\right)$

②
$$\int_0^\infty e^{-st}\left(\frac{\sin t}{t}\right) dt = \tan^{-1}\left(\frac{1}{s}\right)$$

③ $s \to 0$,
$$\int_0^\infty \left(\frac{\sin t}{t}\right) dt = \tan^{-1}(\infty) = \frac{\pi}{2}$$

Pg
$$\int_0^\infty \left(\frac{\sin 2t}{t}\right) dt = \int_0^\infty \frac{\sin 3t}{t} dt = \frac{\pi}{2}$$

Q: Evaluate $L\left\{\frac{\sin at}{t}\right\} = ?$

Sol: W.K.T $L\{\sin at\} = \frac{a}{s^2+a^2} = F(s)$

$$\Rightarrow L\left\{\frac{\sin at}{t}\right\} = \int_s^\infty F(s)\,ds = \int_s^\infty \left(\frac{a}{s^2+a^2}\right) ds$$

$$= a\left[\frac{1}{a}\tan^{-1}\left(\frac{s}{a}\right)\right]_s^\infty = \tan^{-1}\infty - \tan^{-1}\left(\frac{s}{a}\right)$$

$$= \frac{\pi}{2} - \tan^{-1}\left(\frac{s}{a}\right) = \cot^{-1}\left(\frac{s}{a}\right) = \tan^{-1}\left(\frac{a}{s}\right)$$

i.e $\boxed{L\left\{\frac{\sin at}{t}\right\} = \tan^{-1}\left(\frac{a}{s}\right)}$

(*) $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$

(*) $\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$

(*) $\tan^{-1}(x) = \cot^{-1}\left(\frac{1}{x}\right)$

Q: $\int_0^\infty e^{-2t}\left(\frac{\sin 3t}{t}\right) dt = ?$

① $L\left\{\frac{\sin 3t}{t}\right\} = \tan^{-1}\left(\frac{3}{s}\right)$

② $\int_0^\infty e^{-st}\left(\frac{\sin 3t}{t}\right) dt = \tan^{-1}\left(\frac{3}{s}\right)$

③ $s \to 2$, $\int_0^\infty e^{-2t}\left(\frac{\sin 3t}{t}\right) dt = \tan^{-1}\left(\frac{3}{2}\right)$ //

Q: $\int_0^\infty (t^2 \cdot e^{-t} \cdot \sin t)\,dt = ? = \int_0^\infty e^{-t}(t^2 \sin t)\,dt = ?$

Sol: W.K.T, $L\{\sin t\} = \frac{1}{s^2+1} = F(s)$

$\Rightarrow L\{t^2 \sin t\} = (-1)^2 \frac{d^2}{ds^2}(F(s))$

$= \frac{d}{ds}\left[\frac{d}{ds}\left(\frac{1}{s^2+1}\right)\right] = \frac{d}{ds}\left[\frac{-2s}{(s^2+1)^2}\right]$

$= -2\left[\frac{(s^2+1)^2 \cdot (1) - s\left[2(s^2+1)(2s)\right]}{(s^2+1)^4}\right]$

$= -2\left[\frac{1-4s^2}{(s^2+1)^3}\right]$

Now, we have $L\{t^2 \sin t\} = -2\left\{\frac{1-4s^2}{(s^2+1)^3}\right\}$

$\int_0^\infty e^{-st}(t^2 \sin t)\,dt = (\quad)$

$s \to 1$

$\int_0^\infty e^{-t}(t^2 \sin t)\,dt = -2\left[\frac{1-4}{(1+1)^3}\right]$

$= \frac{3}{4}$

Inverse L.T → if $L\{f(t)\} = F(s) \Rightarrow L^{-1}\{F(s)\} = f(t)$

① $\boxed{L^{-1}\left\{\frac{1}{s^n}\right\} = \frac{t^{n-1}}{\lfloor n}}$

eg $L^{-1}\left\{\frac{1}{s^1}\right\} = \frac{t^{1-1}}{\lfloor 1} = 1$

eg $L^{-1}\left\{\frac{1}{s^2}\right\} = \frac{t^{2-1}}{\lfloor 2} = t$

eg $L^{-1}\left\{\frac{1}{s^3}\right\} = \frac{t^{3-1}}{\lfloor 3} = \frac{t^2}{2}$

② $\boxed{L^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}}$

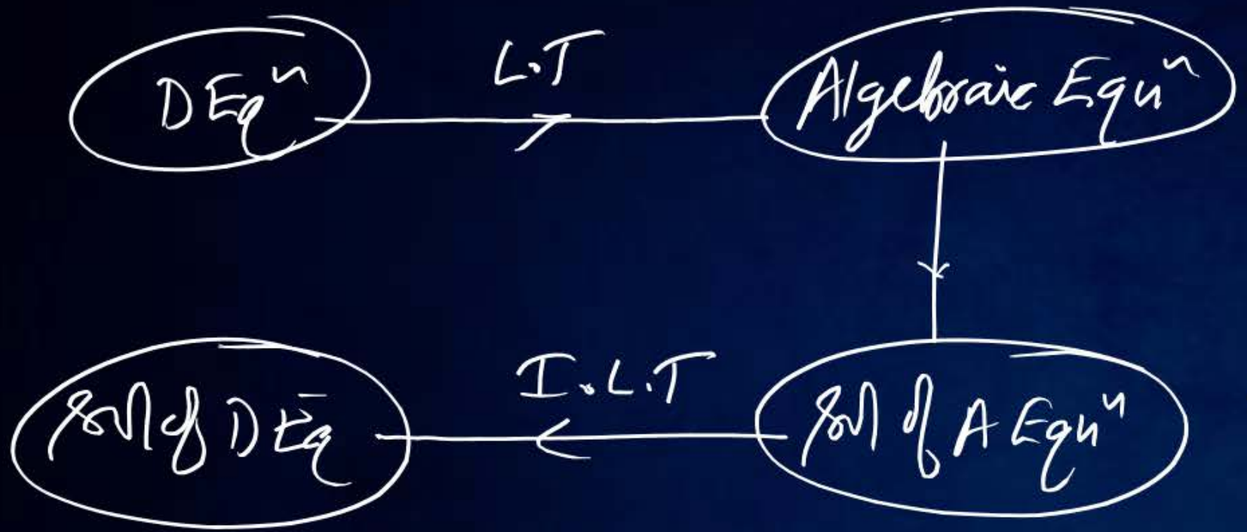
③ $L^{-1}\left\{\frac{1}{s^2+k^2}\right\} = \frac{1}{k}\sin(kt)$

④ $L^{-1}\left\{\frac{1}{s^2-k^2}\right\} = \frac{1}{k}\sinh(kt)$

⑤ $L^{-1}\left\{\frac{s}{s^2+k^2}\right\} = \cos(kt)$

⑥ $L^{-1}\left\{\frac{s}{s^2-k^2}\right\} = \cosh(kt)$

## Application of L.T in the Sol.$^{n}$ of DEq →




Let $y = y(t) = f(t)$ is the Sol$^{n}$ of DEq.

& $Y = Y(s) = F(s)$ is the L.T of Sol$^{n}$.

then $\boxed{L\{y'(t)\} = sY(s) - y(0)}$

or $L\{f'(t)\} = sF(s) - f(0)$

$\boxed{L\{y''(t)\} = s^2Y(s) - sy(0) - y'(0)}$

or $L\{f''(t)\} = s^2F(s) - sf(0) - f'(0)$

& so on - - - - - -

where $\boxed{L\{y\} = Y} \Rightarrow L^{-1}(Y) = y$

Q: Solve $y''(t)+4y(t)=0$; $y=0$ & $\frac{dy}{dt}=3$ at $t=0$ using L.T. technique.

Sol: $L\{y''\}+4L\{y\}=L\{0\}$

$[s^2Y(s)-sy(0)-y'(0)]+4Y(s)=0$

$s^2Y(s)-0-3+4Y(s)=0$

$(s^2+4)Y(s)=3$

or $Y(s)=\frac{3}{s^2+4}$ = L.T of $y(t)$

$L^{-1}\{Y(s)\}=3L^{-1}\left\{\frac{1}{s^2+4}\right\}$

$y(t)=3\cdot\left[\frac{1}{2}\sin 2t\right]$ //

(ii) Find the L.T of the sol of above DEq.

Ans: $Y(s)=\frac{3}{s^2+4}$ //

Q: Solve $\frac{df}{dt} + 3f = e^{2t}$ ; $f(0)=1$ using L.T. Technique.

Sol: Let $f = f(t)$ is the sol & $F = f(s)$ is it's L.T. s.t $\boxed{L\{f(t)\} = F(s)}$

$$f'(t) + 3f(t) = e^{2t}, \quad f(0)=1$$

$$L\{f'(t)\} + 3L\{f(t)\} = L\{e^{2t}\}$$

$$[sf(s) - f(0)] + 3F(s) = \frac{1}{s-2}$$

$$sf(s) - 1 + 3f(s) = \frac{1}{s-2}$$

$$(s+3)f(s) = \frac{1}{s-2} + 1 = \frac{s-1}{s-2}$$

i.e. $f(s) = \frac{s-1}{(s+3)(s-2)} = L.T \text{ of } f(t)$

$$f(s) = \frac{4/5}{s+3} + \frac{1/5}{s-2}$$

$$L^{-1}\{f(s)\} = \frac{4}{5}L^{-1}\left\{\frac{1}{s+3}\right\} + \frac{1}{5}L^{-1}\left\{\frac{1}{s-2}\right\}$$

$$f(t) = \frac{4}{5}(e^{-3t}) + \frac{1}{5}(e^{2t})$$


Thank
YOU